Practice workbook · Arc length, sector area and perimeter

Using s = r theta and A = (1/2) r^2 theta with theta in radians to find an arc, a sector area or a perimeter, or one unknown from a single given fact.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-03

1. Question 1 · part b · 2 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part b only.

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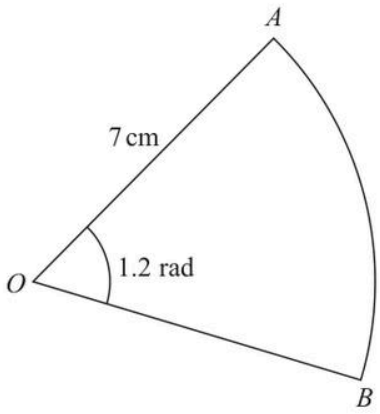


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector AOB of a circle with centre O and radius 7 cm.

The angle AOB is 1.2 radians.

Calculate

(a) the area of sector AOB **(2)**

(b) the perimeter of sector AOB **(2)**

(Total for Question 1 is 4 marks)

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1. Question 1 · part b · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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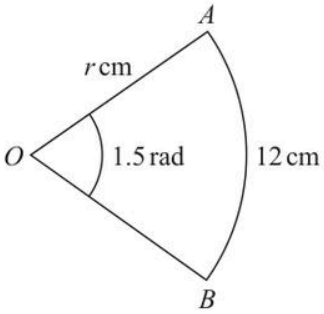


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector AOB of a circle with centre O and radius r cm. The angle AOB is 1.5 radians and the length of arc AB is 12 cm.

Calculate

(a) the value of r,

(1)

(b) the area of the sector AOB.

(2)

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(Total for Question 1 is 3 marks)

2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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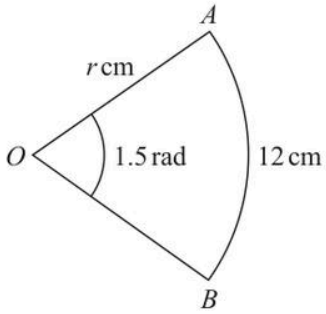


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3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

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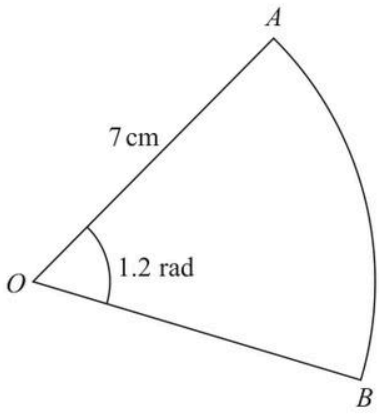


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(b) the perimeter of sector AOB **(2)**

(Total for Question 1 is 4 marks)

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4. Question 1 · part a · 2 marks · Working and answer

Working and answer

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5. Question 3 · part a · 2 marks

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Further practice · Attempt part a only.

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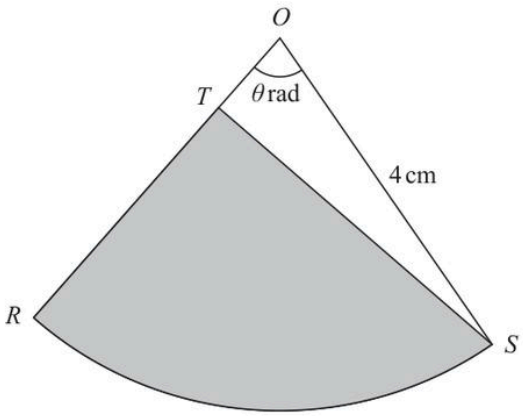


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ORS of a circle with centre O and radius 4 cm.
The size of angle ROS is θ radians.

The area of sector ORS is 2π cm^2

(a) Find the exact value of θ

(2)

(b) Find the perimeter, in cm to 3 significant figures, of the sector ORS.

(2)

The point T lies on OR such that $OT : TR = 1 : 3$

The region shown shaded in Figure 1 is bounded by the line TR, the line TS and the arc RS of the sector.

The area of this region is A cm^2

(c) Find the exact value of A

(2)

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5. Question 3 · part a · 2 marks · Working and answer

Working and answer

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6. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.

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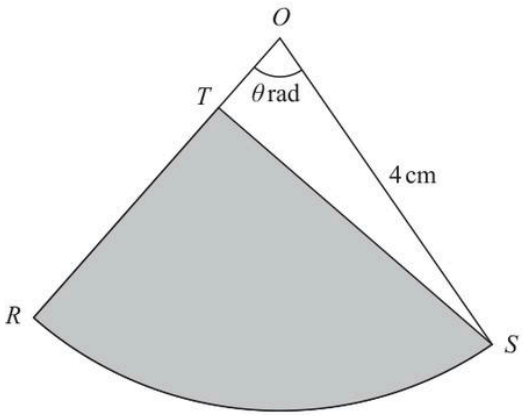


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6. Question 3 · part b · 2 marks · Working and answer

Working and answer

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7. Question 1 · part a · 1 mark

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Further practice · Attempt part a only.

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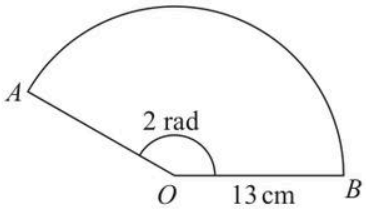


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector AOB of a circle with centre O.
The radius of the circle is 13 cm and angle $AOB = 2$ radians.

(a) Find the length of the arc AB.

(1)

(b) Find the area of the sector AOB.

(2)

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(Total for Question 1 is 3 marks)

7. Question 1 · part a · 1 mark · Working and answer

Working and answer

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8. Question 1 · part b · 2 marks

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Further practice · Attempt part b only.

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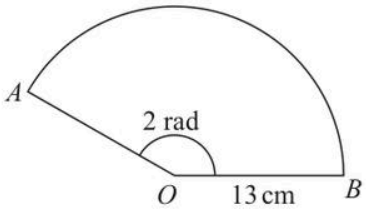


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(a) Find the length of the arc AB.

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(b) Find the area of the sector AOB.

(2)

(Total for Question 1 is 3 marks)

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8. Question 1 · part b · 2 marks · Working and answer

Working and answer

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9. Question 3 · part a · 2 marks

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Further practice · Attempt part a only.

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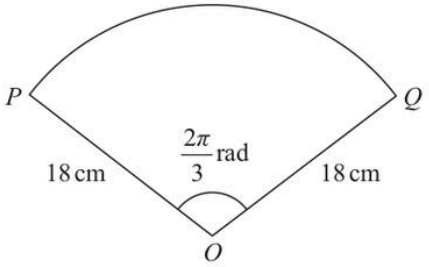


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sector OPQ of a circle with centre O.

The radius of the circle is 18 cm and the angle POQ is $\frac{2\pi}{3}$ radians.

(a) Find the length of the arc PQ, giving your answer as a multiple of π

(2)

Figure 2 below shows the sector OPQ and the kite $OPTQ$.

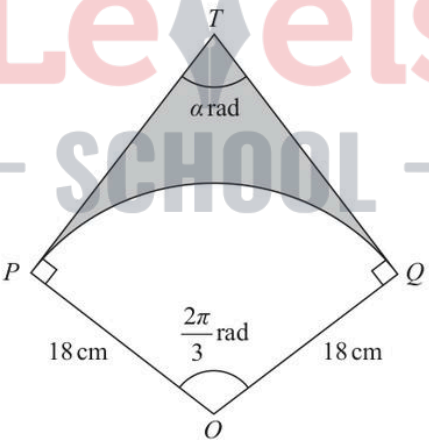


Diagram **NOT** accurately drawn

Figure 2

PT is the tangent to the circle at P and QT is the tangent at Q, such that angle $PTQ = \alpha$ radians.

(b) (i) Find α in terms of π

(1)

(ii) Calculate, to 3 significant figures, the area of the region, shown shaded in Figure 2, which is bounded by the arc PQ and the tangents PT and QT.

(6)

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9. Question 3 · part a · 2 marks · Working and answer

Working and answer

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10. Question 1 · part a · 2 marks

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Further practice · Attempt part a only.

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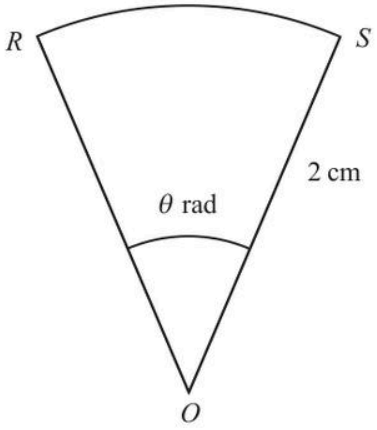


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ROS of a circle with centre O and radius 2 cm
The size of angle ROS is θ radians.

The area of sector ROS is $\frac{\pi}{2}$ cm^2

(a) Find the exact value of θ

(2)

The perimeter of sector ROS is P cm

(b) Find the exact value of P

(3)

(Total for Question 1 is 5 marks)

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10. Question 1 · part a · 2 marks · Working and answer

Working and answer

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11. Question 1 · part b · 3 marks

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Further practice · Attempt part b only.

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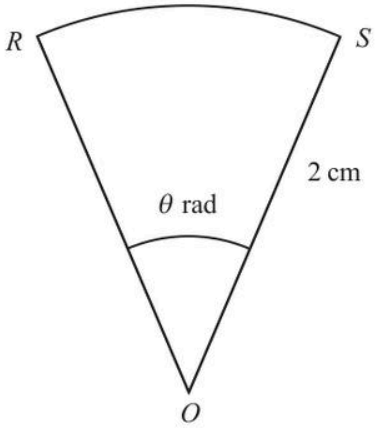


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(3)

(Total for Question 1 is 5 marks)

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11. Question 1 · part b · 3 marks · Working and answer

Working and answer

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